

This was an assignment, which took a lot of interaction with LLM. I was trying to build up on my previous experience from the MARBLE project, where I tried to figure out a way to do phoneme-Level Decoding of pieces of speech from Intracranial EEG under the supervision of Christian Herff and Anirudh Wodeyar, but this specific approach and working with sounds was new to me. I was often asking Claude to bring me up to speed and this is not something I was intending to do from the start, so I do not have a dedicated branch of conversation on this specific project with Claude. However, I asked it to summarize our interaction in this assignment and you can see the snapshot of the answers below. Please note that for some reason it refers to me, Irina Kalmykova, when it says “I” or “me” below.

I hope you would accept this way of disclosing the LLM involvement.

Was it a meaningful discussion?

Yes. Many decisions came from back-and-forth, often with me setting the direction:

- **Permutation scheme:** Claude first proposed permuting labels. I corrected this: "we permuted brain data, not the labels. Our prof was very strict." The above-chance test was rebuilt to permute the brain RDM (rows and columns jointly), and label permutations were kept only for comparing trained vs untrained and different architectures.
- **Multiple comparisons:** I asked: "are you sure bonferroni holm is needed?" We worked out a correction per test: max-statistic across layers for the brain permutation test (layers are dependent), Holm for the run-label tests, the six architecture pairs and the eight depth profiles. We also worked out why the corrected p-values kept landing at the same floor (0.040–0.056). With 5 vs 5 runs there are only 252 label splits, and the floor means complete separation of trained and untrained runs.
- **Metric choice:** Euclidean distance scored higher on YAMNet, and I asked: "should I go for the highest?" We agreed not to, since choosing a metric by its score on the same brain data inflates the result. We kept 1 – correlation on theoretical grounds and reported Euclidean alongside it.
- **Visualization:** the assignment asked for a t-SNE-like visualization. I asked what the options were and chose MDS. When Claude moved ahead to LDA, I said "not done with MDS yet". LDA was added afterwards, to test whether categories are linearly readable when they don't dominate the MDS geometry. I rejected Claude's first category grouping, which led to the six balanced categories (speech, voice, animal, music, nature, tools).
- **Scope:** I dropped the bootstrap (too slow), partial correlations and Hedges' g (overkill).
- **Critical checking of the report:** Claude flagged claims the results didn't support, and we rewrote them:
 - the biological model having "the highest" alignment, when it didn't differ significantly from the inspired model;
 - sound level presented as a strength rather than a confound;
 - "noisy fMRI", when STG actually decoded categories better than the trained models;
 - a citation (Sanghavi 2020) that turned out to be a dataset name, which we replaced with Schrimpf et al.

Where Claude helped with Python

- **RSA core:** 1 — correlation RDMs with NaN handling for constant patterns, and Spearman comparison over the 41,328 sound pairs.
- **Tests:** brain-data permutation with max-statistic correction, run-label permutation with Holm correction, architecture comparisons and depth-slope tests.
- **Descriptive statistics:** Welch confidence intervals, seed consistency, and formatted results tables.
- **Sound-level control:** a log RMS level RDM built from the waveforms.
- **Data handling:** loading the betas with h5py, and parsing filenames into categories and stimulus numbers.
- **MDS:** non-metric MDS with stress and Procrustes alignment to STG, a plotting grid with class colors, percentile color limits for the sound-level map, and PNG export.
- **LDA:** cross-validated LDA (z-scoring, PCA to 30 components, shrinkage LDA, balanced accuracy, held-out projections), with PNG export.

Where Claude helped with LaTeX

- **Tables:** booktabs versions of all results tables, with grouped headers, captions and p_{adj} notation.
- **Compile problems:** Unicode minus, unescaped underscores, `\texttt` overflow, longtable needing two runs.
- **Step-by-step revision:** at my request ("please make sure it is consistent with ipynb and let's go step by step, do not give me back the whole file. Tell me what do I adjust where"), Claude checked the report against the notebook section by section and told me what to change where.
- **Rewrites:** shortened and corrected versions of the MDS, LDA, Discussion, Future Work and Conclusion sections.
- **References:** bibliography items, with advice to verify each one.

Example exchanges

1. Permutation test

- *Me*: "We permuted brain data, not the labels. Our prof was very strict."
- *Claude*: Rebuilt the above-chance test to shuffle which brain pattern belongs to which sound, by jointly permuting the rows and columns of the brain RDM. It used the same permutations for all layers and a max-statistic correction.

2. Multiple comparisons

- *Me*: "Are you sure Bonferroni-Holm is needed?"
- *Claude*: Testing every layer inflates false positives, so a correction is needed. Across dependent layers of one network, the max-statistic method is better suited. Holm fits the run-label tests.

3. Metric choice

- *Me*: "Should I go for the highest?"
- *Claude*: No. Selecting the metric by its score on the same brain data is circular. Choose on principle and report the other metric for transparency.

4. Sound-level figure

- *Me*: After the percentile color fix: "It got better, right?"
- *Claude*: Yes. The quiet sounds were now visible, clustering in the first layers of all models and moving to distinct positions deeper in, most clearly at the biological model's A4 stage.

Please create a summary of our dialogues related to Assignment 4 (RSA analysis of different setups for auditory processing) explaining how I interacted with you . What kind of questions did I ask, whether we engaged in a meaningful discussion, where you helped me with the python code and latex. Please provide a few examples of questions and answers from this thread. Please also mention what came from my previous knowledge and experience of working in projects related to studying the brain activity.

Working with Claude on Assignment 4: RSA of auditory models vs STG fMRI

Context

Assignment 4 of Multiscale Modeling of Biological Systems: Neuroscience was a group report. I was responsible for Part I: representational similarity analysis (RSA) comparing the provided auditory networks (waveform, uninspired, inspired), our group's biological model and YAMNet with single-participant STG fMRI responses to 288 natural sounds. I used Claude throughout, for the analysis notebook, the statistical design, the figures and the LaTeX report. I led the work. I brought the assignment text, my notebook, outputs, error messages, figures and the team's draft, and I made the decisions about what to analyze and what to leave out.

Kinds of questions I asked

- **Understanding the setup:** what the provided code does, what exactly is being modeled, how voxel responses and the fMRI (BOLD) signal relate to model activations, what STG is, whether the models were trained on brain data, and whether the STG data covered one hemisphere or both.
- **Methodological choices:** Pearson vs Spearman, Euclidean vs 1 — correlation, which permutation scheme to use, whether Bonferroni-Holm is needed, max-statistic vs Holm, which visualization to use, whether PCA belongs inside LDA, and whether an effect size (Hedges' g) adds anything.
- **Implementation and debugging:** code cells for each analysis, and fixes for errors such as undefined variables, a MATLAB v7.3 file that scipy couldn't read, filename regex mismatches, sklearn warnings and plot layouts.
- **Writing and checking:** shortening sections, updating text to new figures, checking claims against the results ("is it fair to call the waveform model the worst?"), and finding references.
- **LaTeX:** table layout, compile errors, captions, notation for adjusted p-values, and text overflowing the margin.

What came from my prior experience with brain-activity research

Before this course I worked on the MARBLE project at Maastricht University: phoneme-level decoding from intracranial EEG for neural speech prostheses, built on the SingleWordProductionDutch dataset and supervised by Christian Herff and Anirudh Wodeyar. That background shaped the assignment in several ways:

- **Knowing what the brain signal is:** from iEEG I was used to fast electrophysiological signals. I made a point of clarifying how fMRI differs: betas per voxel, a slow hemodynamic signal, one measurement per sound. Those questions fed into the limitations (no noise ceiling) and the Future Work point that spiking models may show their benefit more in electrophysiology than in fMRI.
- **Decoding methods:** my speech-decoding pipelines included LDA-based classification, so cross-validated, held-out LDA with chance-level reasoning was familiar ground. It became the natural complement to MDS.
- **Speech-specific interest:** from my speech-prosthetics work I was interested in how the brain treats human speech compared with other sounds. This led to keeping speech and voice as separate categories, to the finding that STG separates human from animal and music sounds, and to my Future Work idea of analyzing the left and right auditory cortex separately.
- **Research practice:** experience with small single-participant datasets made me cautious about over-interpreting significance, which matched the careful framing we kept in the report.

Claude's record of the early part of our conversation was condensed, so exchanges 1-3 are quoted from my messages, but the answers are paraphrased. The section on prior experience describes how my background influenced the work; I didn't necessarily state each of these points explicitly in the conversation. Adjust the wording where it differs from how you remember it.

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